Case ID	Test Category	Test Description	Test Query	Expected Result	Comment
TC- 01	**Procedure**	Add a standard exam for an authorized instructor	`EXEC USP_AddExam @courseId=1, @examType='Exam', @intakeTrackId=10, @startTime='2026-05-01', @endTime='2026-05-02';`	Success	Verified instructor belongs to the class.
TC- 02	**Procedure**	Add 'Corrective' exam without a previous 'Exam'	`EXEC USP_AddExam @courseId=2, @examType='Corrective', @intakeTrackId=10, @startTime='2026-06-01', @endTime='2026-06-02';`	Error	Logic prevents corrective exams without a base exam.
TC- 03	**Procedure**	Unauthorized instructor creating an exam	`EXEC USP_AddExam @courseId=99, @examType='Exam', @intakeTrackId=10, ...`	Error	"Not authorized to create an exam for this class."
TC- 04	**Procedure**	Fill exam with random MCQs	`EXEC USP_AddRandomQuestionsToExam @examId=1, @mcqCount=5, @mcqMarks=2, @tfCount=0, @textCount=0;`	Success	5 random MCQs inserted into `ExamQuestion`.
TC- 05	**Procedure**	Request more MCQs than available in bank	`EXEC USP_AddRandomQuestionsToExam @examId=1, @mcqCount=1000, ...`	Error	"Insufficient MCQ questions in the bank."
TC- 06	**Trigger**	Insert a duplicate choice for a question	`INSERT INTO Choice (QuestionId, ChoiceText) VALUES (10, 'Option A');`	Error	`TR_PreventDuplicateChoice` fired successfully.
TC- 07	**Trigger**	Update a class to a combination that exists	`UPDATE Class SET CourseId=1, InstructorId=5 WHERE ClassId=10;`	Error	`TR_PreventDuplicateClass_Update` blocked the action.
TC- 08	**Trigger**	Unauthorized user modifying `Branch`	`SET SESSION_CONTEXT(N'UserId', 999); INSERT INTO Branch ...`	Error	`TR_CheckPermission_Branch` access denied.
TC- 0G	**Trigger**	Admin/Instructor modifying `Intake`	`SET SESSION_CONTEXT(N'UserId', 1); INSERT INTO Intake ...`	Success	Permission granted for high-level roles.
TC- 10	**Trigger**	Submit answer after exam `EndTime`	`INSERT INTO StudentAnswer (StudentExamId, QuestionId, Answer) ...`	Error	`TR_StudentAnswer_CheckExamTime` blocked submission.

TC- 11	**Procedure**	Delete instructor assigned to classes	`EXEC USP_DeleteInstructor @InstructorId = 5;`	**Error**	Prevents deleting active instructors.
TC- 12	**Procedure**	Successfully delete a free instructor	`EXEC USP_DeleteInstructor @InstructorId = 90;`	**Success**	Instructor, User, and Login dropped correctly.
TC- 13	**Function**	Check if a specific question exists	`SELECT dbo.FN_CheckQuestionExists('What is C#?');`	**1**	Returns 1 (True) if question body is found.
TC- 14	**Function**	Check if a class exists	`SELECT dbo.FN_CheckClassExists(5, 1,10);`	**1 / 0**	Validates Instructor-Course-Track link.
TC- 15	**Table Function**	Fetch exam paper for instructor	`SELECT * FROM dbo.TVF_GetExamPaperForInstructor(1, 'Exam');`	**Table Results**	Displays questions with aggregated choices.
TC- 16	**Procedure**	Update course with partial data (NULLs)	`EXEC USP_UpdateCourse @CourseId=31, @MaxDegree=100;`	**Success**	`ISNULL` logic kept other fields unchanged.
TC- 17	**Procedure**	Delete a branch	`EXEC USP_DeleteBranch @BranchId = 2;`	**Success**	Row removed from `Branch` table.
TC- 18	**Procedure**	Remove track from intake	`EXEC USP_RemoveTrackFromIntakeAtBranch @IntakeTrackId = 5;`	**Success**	Row removed from `IntakeTrack`.
TC- 1G	**Procedure**	Update exam start/end time	`EXEC USP_UpdateExam @examId=1, @startTime='2026-07-01';`	**Success**	Time updated with instructor authorization check.
TC- 20	**Audit Check**	Verify Audit log after an Error	`SELECT TOP 1 * FROM AuditTable ORDER BY ChangeDate DESC;`	**Data Row**	Confirms `CATCH` blocks are logging errors.
TC- 21	**Constraint**	Add Exam with StartTime > EndTime	`EXEC USP_AddExam ..., @startTime='2026-10-01', @endTime='2026-09-01';`	**Error**	DB Constraint prevents illogical dates.
TC- 22	**Procedure**	Add New Branch that already exists	`EXEC USP_AddNewBranch 'Cairo', 'Maadi';`	**Message**	"Adding Cannot Complete, the Branch Exists!"
TC- 23	**Trigger**	Prevent duplicate Class insertion	`INSERT INTO Class (InstructorId, CourseId, IntakeTrackId) ...`	**Error**	`TR_PreventDuplicateClass` blocked the insert.
TC- 24	**View**	View all instructor details	`SELECT * FROM All_Instructor_Data;` (Based on your imgs)	**Success**	Returns joined data from User and Instructor.
TC- 25	**Function**	Get Instructor total classes count	`SELECT dbo.Get_Instructor_Class_Count(5);`	**Numeric Value**	Returns correct count of assigned classes.